Exp. No: 5

Electric Potential Difference between Two Points in Free Space

Aim:

To write program for Electric potential difference between two points in free space medium using MATLAB.

Objectives:

1. To determine the electric potential difference between two points separated by a distance R in free space.

2. To plot the variation of potential difference with separation distance.

3. To relate the potential gradient to electric field intensity for validation.

PROGRAM

```
clc;
clear;
q = input('Enter the q value: ');
epsilon = 8.854e-12;
k = 1/(4*pi*epsilon);

ax = input('Enter the value of ax = ');
ay = input('Enter the value of ay = ');
r1 = sqrt(ax^2 + ay^2);
v1 = k*q/r1;

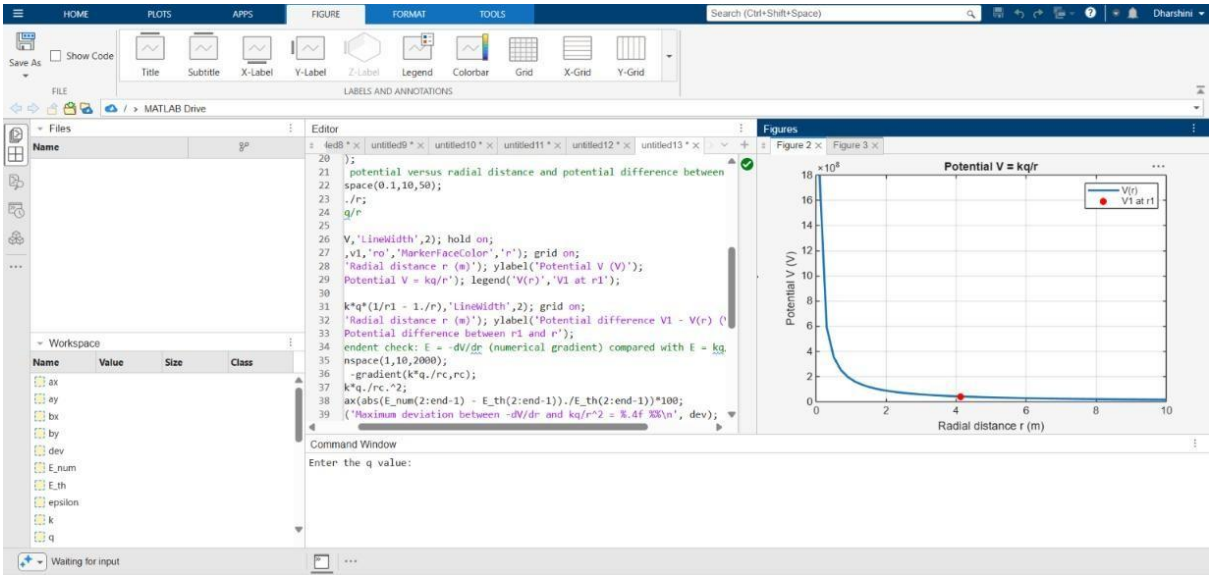
disp('The value of v1');
disp(v1);

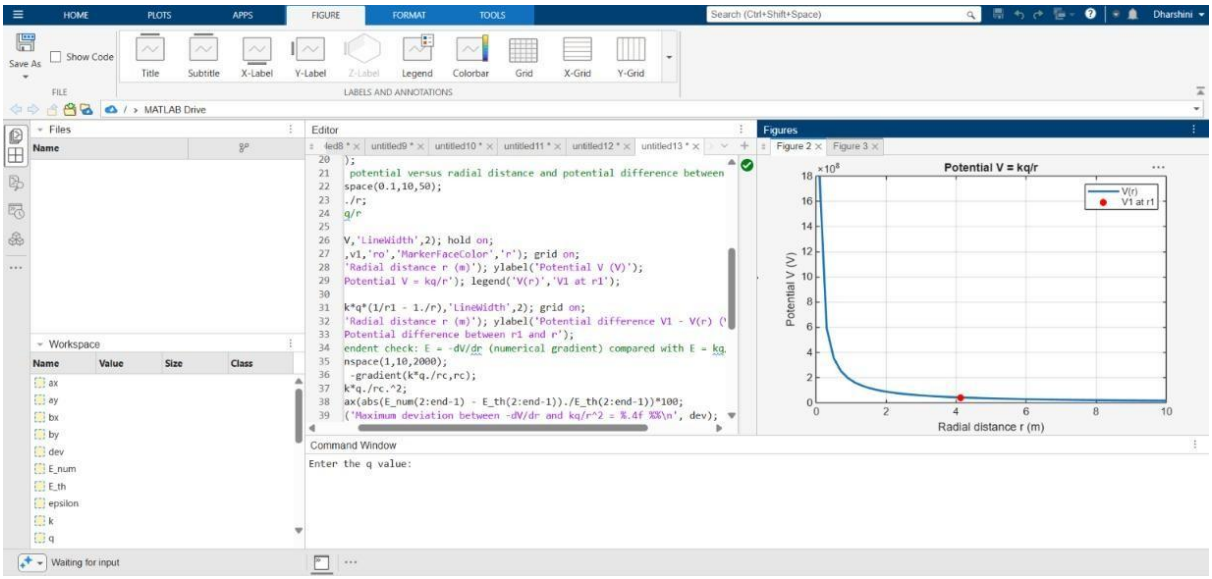
bx = input('Enter the value of bx = ');
by = input('Enter the value of by = ');
r2 = sqrt(bx^2 + by^2);
v2 = k*q/r2;

disp('The value of v2');
disp(v2);

v = v1-v2;
disp('The electric potential difference is');
disp(v);
```

OUTPUT





5(b) Potential Difference for a Positive Charge with Fixed \(r_1\) and Varying \(r_2\)

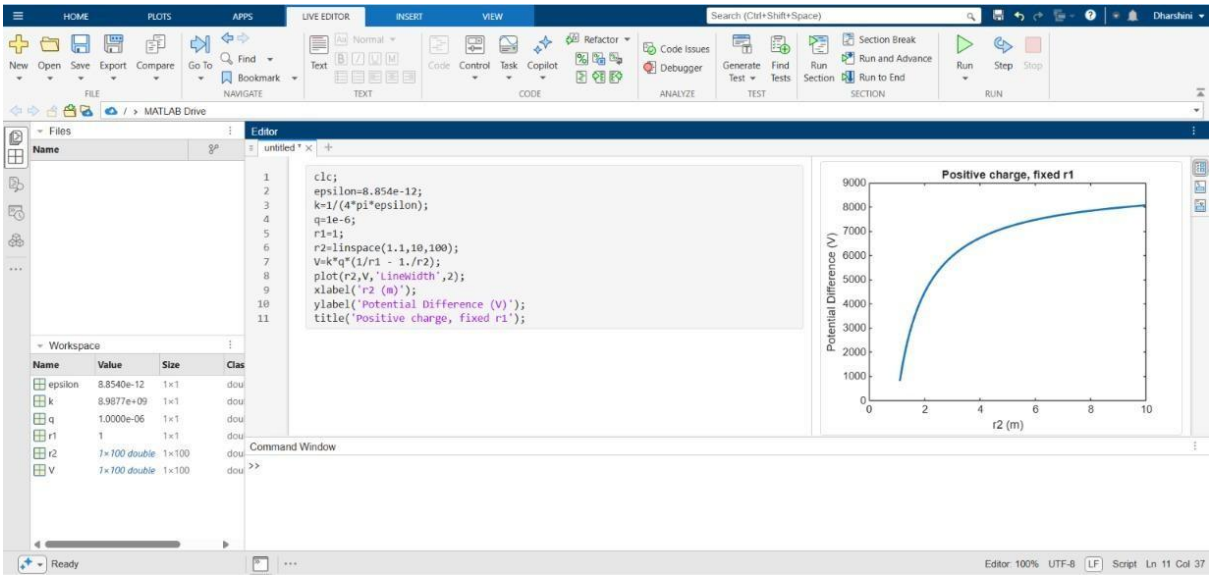
```
clc;
clear;
epsilon = 8.854e-12;
k = 1/(4*pi*epsilon);
q = 1e-6;
r1 = 1;
r2 = linspace(1.1,10,100);

V = k*q*(1/r1 - 1./r2);

plot(r2,V,'LineWidth',2);
grid on;
```

```
xlabel('r2 (m)');
ylabel('Potential Difference (V)');
title('Positive charge, fixed r1');
```

OUTPUT



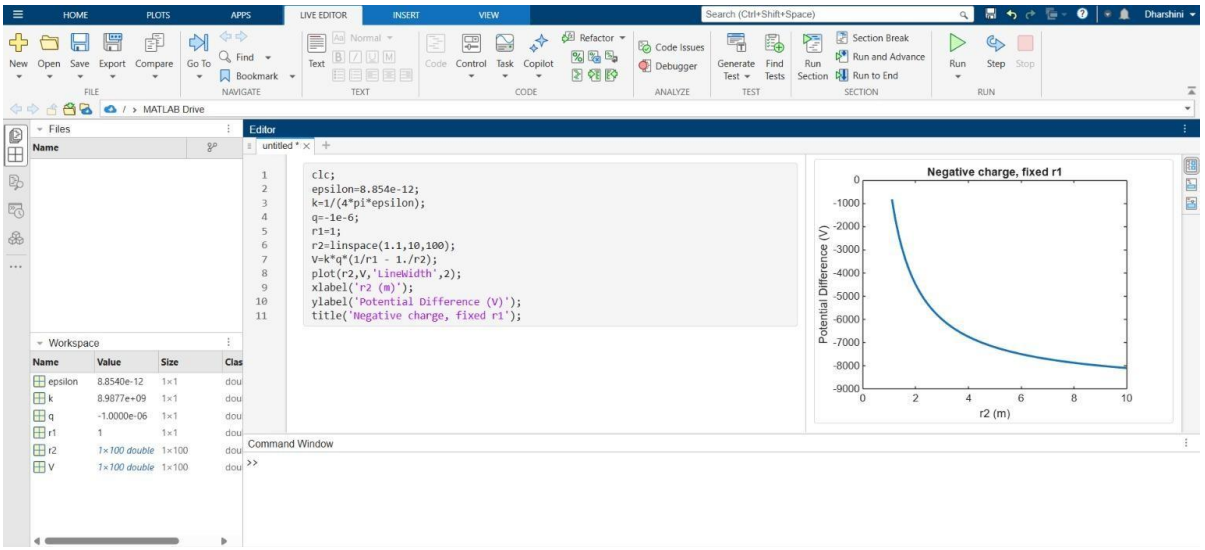
5(c) Potential Difference for a Negative Charge

```
clc;
clear;
epsilon = 8.854e-12;
k = 1/(4*pi*epsilon);
q = -1e-6;
r1 = 1;
r2 = linspace(1.1,10,100);

V = k*q*(1/r1 - 1./r2);

plot(r2,V,'LineWidth',2);
grid on;
xlabel('r2 (m)');
ylabel('Potential Difference (V)');
title('Negative charge, fixed r1');
```

OUTPUT



5(d) Verification of Electric Field Using Potential Gradient

```
clc;
clear;
epsilon = 8.854e-12;
k = 1/(4*pi*epsilon);
q = 1e-6;

r = linspace(1,10,2000);
V = k*q./r;

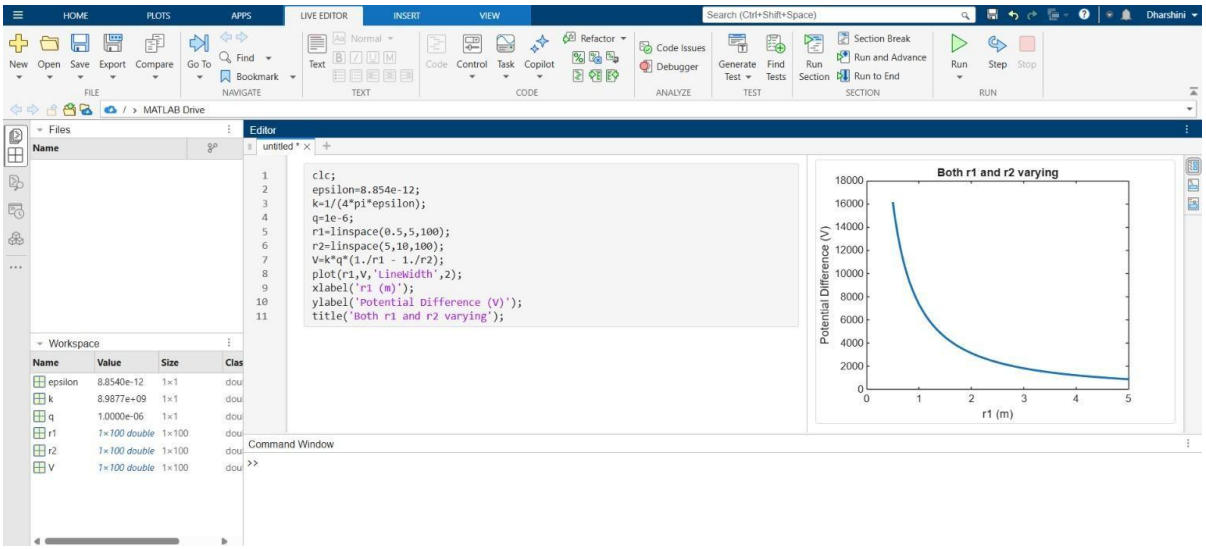
E_num = -gradient(V,r);
E_th = k*q./r.^2;

plot(r,E_num,'b','LineWidth',2);
hold on;
plot(r,E_th,'r--','LineWidth',2);
grid on;
xlabel('Distance (m)');
ylabel('Electric Field (V/m)');
title('Electric Field Verification');
legend('Numerical E','Theoretical E');

dev = max(abs(E_num(2:end-1)-E_th(2:end-1)) ...
    ./abs(E_th(2:end-1)))*100;

fprintf('Maximum deviation = %.4f %%\n',dev);
```

OUPUT



S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)	Marks given	Marks obtained
1	Define charge, reference and observatio n points	Define charge, reference and observation points	Correctly identifies charge distribution, point locations, distance and reference convention.	Setup is correct with minor clarifica tion needed.	Some input or reference details are unclear.	Uses an incorrect setup or reference.	25	2.5	
2	Calculate potential and potential difference	Calculate potential and potential difference	Calculates potential at both points and signed difference correctly with units.	Results are mostly correct with minor arithmet ic errors.	Partial calculations require support.	Potential calculations are incorrect or absent.	25	2.5	
3	Verify potential relationshi p	Verify potential relationship	Checks results against analytical equations and, where relevant, the relation between E and potential.	Validati on is mostly correct with minor omissio ns.	Validation is incomplete.	No valid theoretical check is provided.	30	3	

4	Interpret and report findings	Interpret and report findings	Explains the sign and physical meaning of potential difference with clear tables/conclusion.	Explanation is correct with minor omissions.	Explanation is limited or unclear.	Cannot explain or document the result.	20	2	
Total							100	10	